

CS422 Lab4: A Synthesis Example via Wireshark

Due: **23:59:59 PM, Fri Dec 05, 2025**

Total points: **30 points**

Goal

In this lab, you are going to use **Wireshark** to capture data packets while you use WiFi to do a synthesis example illustrated in the lecture slide and write some codes to analyze the packets captured by you. You should use Python to write the codes to offline analyze packets captured by **Wireshark**.

1 Programming Assignments and Questions

1.1 Part A (5 points)

Please use Wireshark to capture packets over WiFi.

- First of all, please make sure that you use **PAL3.0** (our campus wireless network), not Ethernet. Please close all the browsers and apps that use the Internet as much as you can, which is not mandatory but recommended as it is helpful to reduce packets captured by **Wireshark**.
- Then, please start packet capturing via **Wireshark**.
 - Please first turn off WiFi and then quickly turn on WiFi.
 - Once WiFi (PAL3.0) is on, please quickly open a web browser (say, Chrome or Firefox or Safari or Edge) and go to **www.youtube.com**.
 - At the Youtube website, please search “TED”.
 - When you receive all the returned results, please click one Ted talk to watch.
 - Please watch this video for a few seconds (it is fine to watch it longer but there is no need of doing so). Now you can stop packet capturing.
- Please save the captured pcap file as **lab4.pcap**.

1.2 Part B (15 points)

Please write your own code **lab4.py** to analyze the captured pcap file and extract the packets. Please use **Python3**. Please run packet analysis to handle the following three cases:

- **CASE A:** Please extract and analyze all the packets used for IP address allocation. Please output all the packets IN ORDER which are used to assign the IP address to your computer. Please finally print the IP address of your computer and the IP address of the gateway router.
Note: Wireshark may capture packets that assign IP address for other computers over the same WiFi network. Your code should handle this case and figure out what packets are used for your own IP address allocation.
- **CASE B:** Please extract and analyze all the packets used to get the MAC address of the gateway router. Please print the whole packet (in HEX), as well as the IP address and the MAC address of the gateway router.
Note: Case B needs the output of Case A.

- **CASE C:** Please extract all the packets used to get the IP address of the destination website (here, youtube.com). Please print the IP address of the destination website. *Note: there are multiple IP address for the destination website and you only need to print the first IP address.*

You are recommended to use only **scapy** library. However, if you need other libraries, please provide a script (**install.sh**) to install all needed python3 lib so that your python3 code can run.

Your code should support every case (A, B, C) and another mode “ALL” which run all the cases. To run your code, please use the following command

```
python3 lab4.py [CASE] [Input pcap file] [Destination website]
```

- [CASE] is optional. It is by default ALL and it also accepts A, B or C.
- [Input pcap file] is mandatory. You can use the file captured by you (lab4.pcap), or the test file used by lab4 (lab4-test.pcap).
- [Destination website] is mandatory for Case C and ALL. Note that we will change the destination website and pcap file when we use other test cases to grade your codes.

Note that if there are more than one packet, please get them IN ORDER.

Your code should print out the results in the following format where the first line starts with CASE NO (say, CASE A, CASE B, CASE C), the following lines followed by the results. To print one packet, please just print its Frame No. For other information, please start with a header name, followed by “: ”, and the header value. Note there is a space after “:”. For example, when the mode is set to “ALL”, the sample output is as below, please strictly follow the semantics of the output:

```
CASE A:
Frame 70
Frame 80
Frame 81
Frame 82
Frame 84
Frame 85
IPAddr-computer:  10.184.98.157
IPAddr-gateway:   10.184.0.1

CASE B:
Frame 95
Frame 96
IPAddr-gateway:   10.184.0.1
MACAddr-gateway:  00:00:0c:9f:f0:01

CASE C:
Frame 3566
Frame 3572
name:  www.youtube.com
IPAddr:  108.177.122.136
```

Note: the values in the above example can be different from yours. The test cases used to grade your code are different from the sample case (lab4-test.pcap).

1.3 Part C (10 points)

Please use your codes and Wireshark to process the sample pcap file (lab4-test.pcap).

1. (2 points) In the sample pcap file, what protocols are used to IP address and the MAC address of the gateway router? Which address do you get first?
2. (2 points) In the sample pcap file, have you obtained the MAC address of other hosts (different from the IP address of the gateway router)? If yes, please list all these IP addresses. If no, please justify why.
3. (2 points) In the sample pcap file, what is the CNAME of the destination website (www.youtube.com)? How many IP addresses have been found for the destination website?
4. (2 points) In the sample pcap file, does it use TCP or UDP to visit the destination website (www.youtube.com)? Why?
5. (2 points) Please use IP lookup service (say, <https://whatismyipaddress.com/ip-lookup>) to figure out where is the IP address of the destination website? In this sample pcap file, does it use the destination IP address which is physically closer to you?

2 Materials to turn in

Please submit your assignment on **Gradescope**. You should submit a zip file named as “lab4-UID*.zip” including

- lab4.pcap. If the file is larger than 10MB, please create a new pcap file called lab4.pcap after you remove those packets before the first DHCP message and those packets with the frame number larger than 1000+ the first packet to the designation website.
- the source code files (lab4.py). Please include install.sh if applicable. Please include a readme on how to compile and run your source code (if not exactly the same as our lab convention where we run python3 lab4.py). It is not recommended, but please include it if needed.
- lab4.pdf, a lab report that shows the output from Part B and answers the questions listed in Part C. Please include your name and student ID at the top of the first page.

3 More tips and Support

1. If you cannot use PAL3.0, you can test with your own WiFi network. Please check whether IPV4 or IPV6 is used. We accept both, but please make a note at the start of your report if you do not use PAL3.0 and if this WiFi networks uses IPv4 or IPv6.
2. If you want to use any late day, please send an email to cs422-ta@cs.purdue.edu and tell us. Please do not forget that each student can only use up to 3 late days for all the assignments.
3. Questions about the assignment should be posted on **Campuswire** or asked during **PSOs** or office hours.